

Quorum before execution · KAVACHA before damage · x402 inside the agent economy.



A production, live agentic system at the convergence of Agentic AI × DeFi × RWA on Casper.

Casper ID **martian3062**

Self-Healing Agentic DeFi Treasury Swarm

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Live: https://35.255.196.78.nip.io/eraya_api/demo/

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Contents

Section	Slides
Overview & problem	1–3
How it works	4–6
Casper & AI stack	7–8
Security & live proof	9–11
Why ERAYA & the ask	12–15

DeFi does not fail politely.

Humans and single bot's break

Treasuries are run by humans or single “trade bots” — both both break under adversarial, real-money conditions.

Failure paths are not optional

On-chain systems punish missing failure paths with irreversible loss, not just a bad log line.

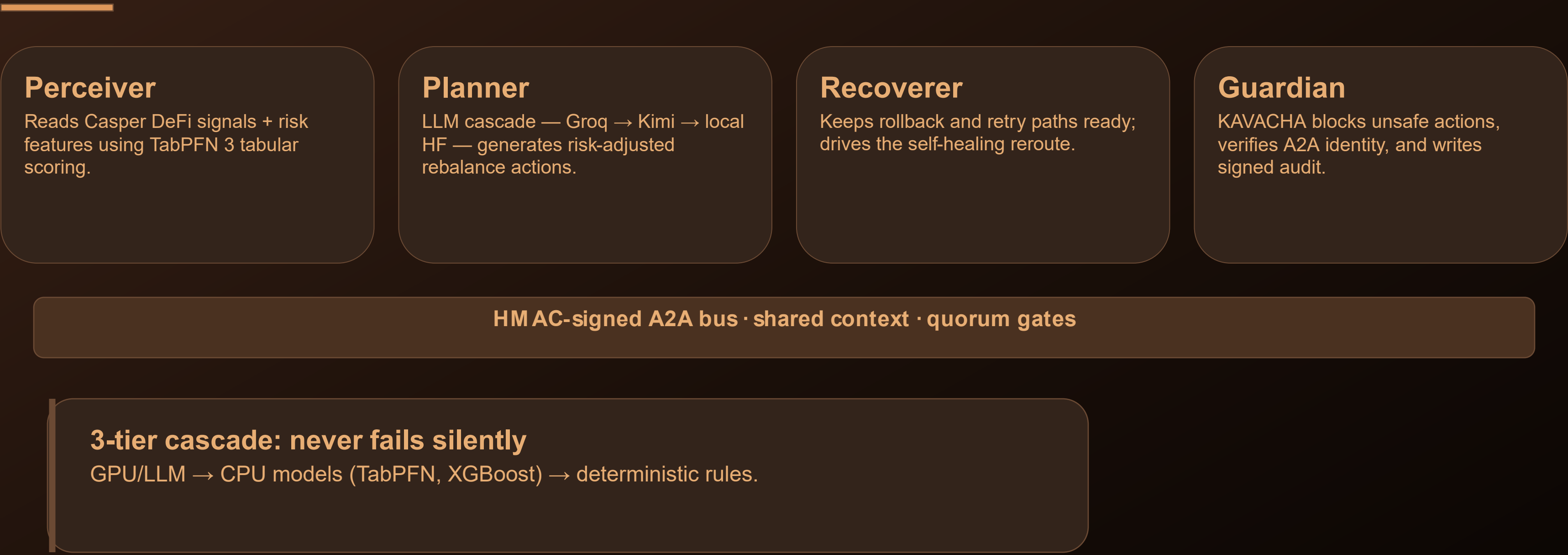
Happy-path demos are not enough

Most agent demos optimise the happy path; none prove what happens when a fix itself fails.

Identity and review are missing

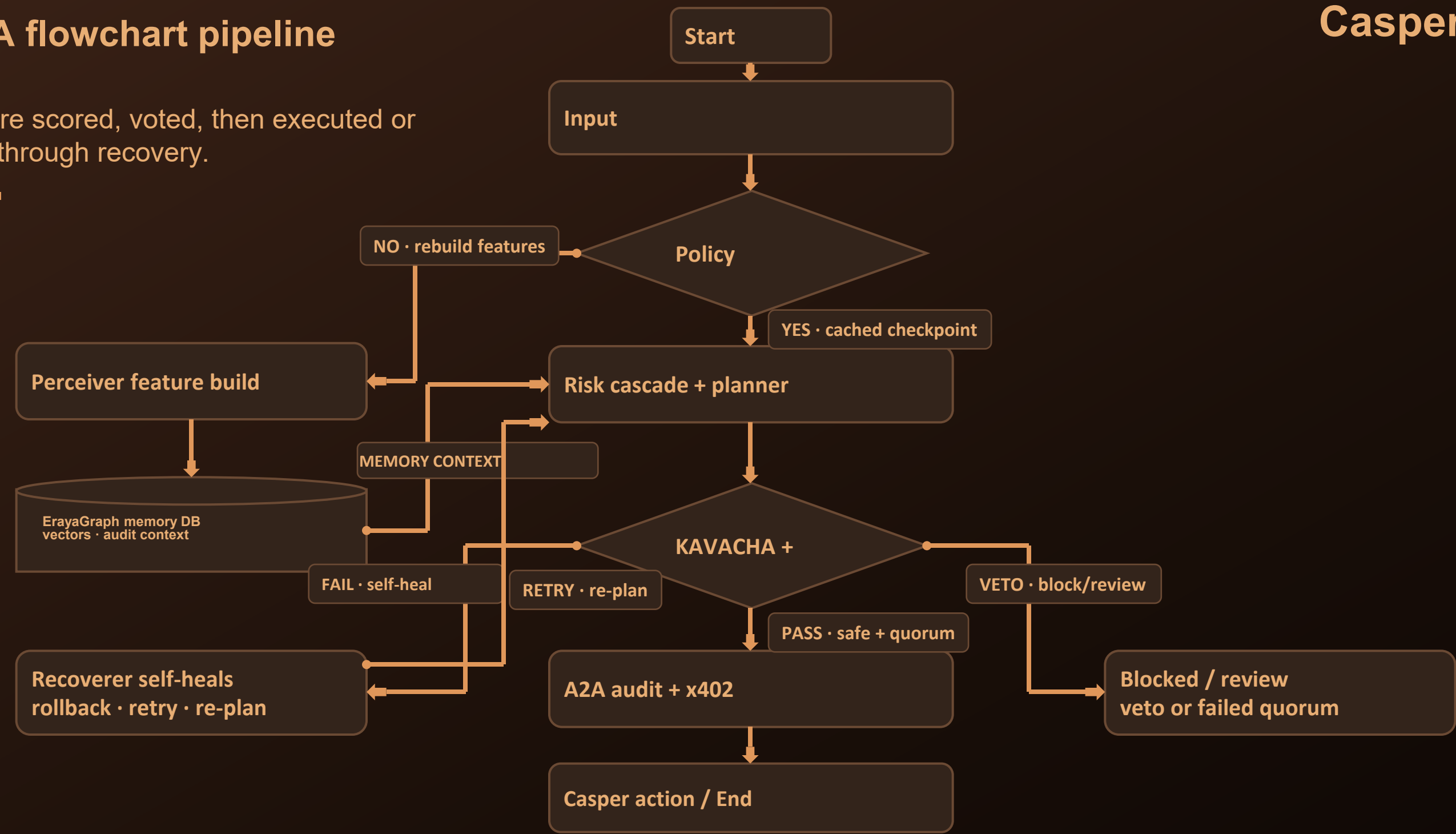
No on-chain agent identity, no economic signal for useful agent work, no adversarial review before high-stakes moves.

Perceiver → Planner → Recoverer → Guardian, over an A2A bus.



ERAYA flowchart pipeline

Signals are scored, voted, then executed or rerouted through recovery.



HMAC-signed A2A bus · shared memory · quorum gates · x402-priced agent work

Casper-native, now writing to the chain.

casper_defi domain adapter

Live CSPR.cloud / CSPR.trade-shaped telemetry; the live validator set feeds decentralization detection.

pool TVL APY gas slippage liquidity depth governance quorum

validator uptime mempool

x402 payments (LIVE)

HTTP-402 challenge → pay
→ 200 flow served for paid
market data.

CSPR.click + MCP

CSPR.click wallet + Casper
MCP facades — graceful
demo→live switch.

Odra contracts (next)

Agent registry, treasury,
governance, and reputation
on Casper testnet.

On-chain anchoring

Critical anomalies anchored
as real Casper 2.0 testnet
transactions via casper-client
put-transaction — evidence
hash verifiable on
testnet.cspr.live.

Threats hunted before a bad deploy lands — and anchored on-chain.

KAVACHA

Security-first guardian: blocks unsafe actions and injection, verifies agent identity, and anchors critical findings on-chain.

Critic agent (5th reviewer)

Independent LLM-as-judge scores every high-stakes action
APPROVE / REVISE / REJECT before execution.

Injection kill-shot

Unsafe inputs BLOCKED with a signed audit trail — live: rule R003, full timeline.

A2A identity defense

Forged agent messages rejected via HMAC — same function as the live path.

Quorum before execution

High-stakes moves need swarm consensus, not one agent's call.

Network Anomaly Intelligence

Statistical detection over mempool, governance & validator telemetry — z-score baselines, MEV correlation, Sybil clustering (Gini/entropy), Nakamoto coefficient; findings anchored on Casper testnet.

Defined failure behaviour is our moat.

Every agent has a defined failure path

The swarm never fully stops — it can re-plan, retry, roll back, or fail safely.

Actually on-chain

Anomalies are anchored as real, verifiable Casper testnet transactions — not "coming soon."

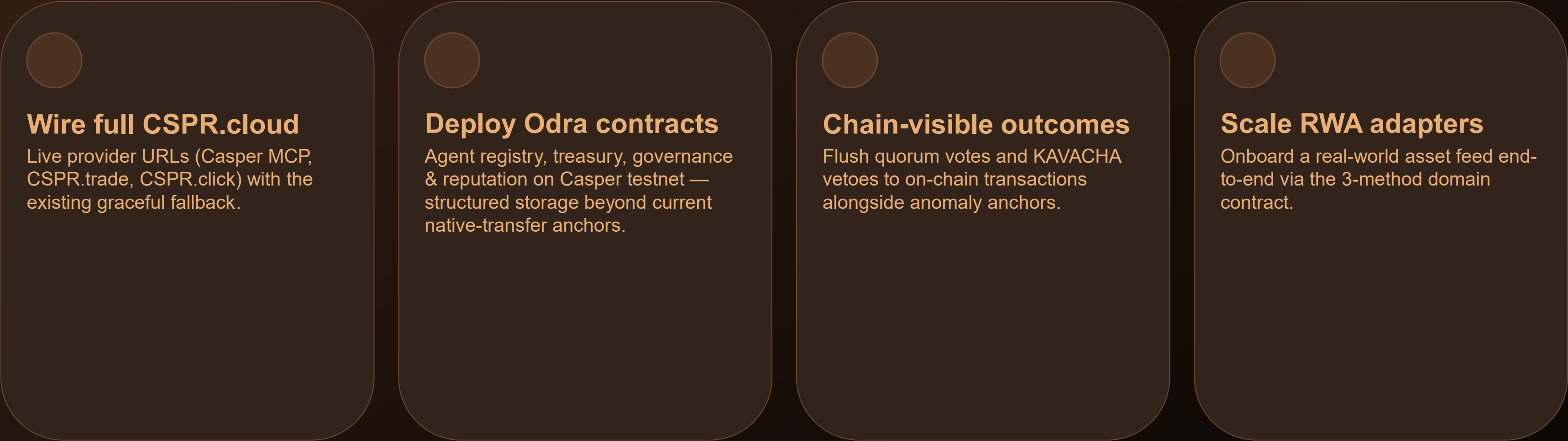
Real signals on the critical path

Real LLM planning + real TabPFN risk + real on-chain writes + real x402 payments — no stubs on the critical path.

Two hard problems solved together

Agent safety (KAVACHA + statistical anomaly intel + Critic) + agent economics (x402 + on-chain anchoring).

ERAYA Casper on CSPR.fans to reach the Final Round.



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